Randomized and low-rank approximation

Optimal pivot count for RPCholesky trace approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because a uniform near-optimal pivot count must exploit adaptive residual structure; community impact is efficient kernel and PSD matrix approximation.
Topic: randomized low-rank approximation; kernel matrices

Context and notation

For a Hermitian positive-semidefinite $A \in \mathbb{C}^{n\times n}$, define the exact-arithmetic RPCholesky residuals by $R_0 = A$. Conditional on $R_t \neq 0$, select j with probability $(R_t)_{jj}/\operatorname{tr}(R_t)$ and set

$$R_{t+1} = R_t - \frac{R_t(:,j)R_t(j,:)}{(R_t)_{jj}}.$$

If $R_t = 0$, keep all subsequent residuals zero. Eigenvalues are ordered $\lambda_1(A) \geq \cdots \geq \lambda_n(A) \geq 0$, and $\tau_r(A) = \sum_{j>r} \lambda_j(A)$. This is the pivot rule in Chen, Epperly, Tropp, and Webber, *Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations*, Algorithm 1; their Lemma 5.5 gives the current comparison $\mathbb{E}\operatorname{tr}(R_r) \leq 2^r \tau_r(A)$.

Problem statement

Does there exist a universal constant $C \geq 1$ such that, for every $n \geq 1$, every Hermitian positive-semidefinite $A \in \mathbb{C}^{n\times n}$, every integer $1 \leq r \leq n$, and every $0 < \varepsilon < 1$, RPCholesky satisfies

$$\mathbb{E}\operatorname{tr}(R_k) \leq (1+\varepsilon)\tau_r(A), \qquad k = \min\{n, \lceil Cr/\varepsilon \rceil\}?$$

The expectation is over the algorithm's adaptive pivots. The constant must be independent of dimension, spectrum, rank, and tolerance. The target concerns this fixed pivoting rule.

References

Epperly, *A new analysis of the randomly pivoted Cholesky algorithm*, §1.2, conjecture immediately after Eq. (1.5), and Corollary 1.3. Earlier motivation appears in Epperly, *Make the Most of What You Have*, Caltech dissertation (2025), §11.1, p. 181.

Status check

The August 2026 paper proves the larger bound $k \geq r/\varepsilon + 2r\sqrt{\log r} + r\log(1/\varepsilon) + 2.3r$ and explicitly conjectures the displayed improvement. Searches for the paper's title, *RPCholesky optimal r epsilon*, and *RPCholesky conjecture* found no subsequent resolution. The arXiv record listed only v1, posted August 21, 2026. The weaker dissertation Conjecture 11.1 is therefore excluded as resolved by this later preprint.

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Rechecked the August 2026 conjecture after (1.5) and its current arXiv record, still v1. Optimal-pivot and RPCholesky follow-up searches found no resolution. The proved extra rank-dependent term does not establish the uniform $O(r/\varepsilon)$ target.